

Fan Yang

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Research Profile

Computational Engineering Master's student specializing in numerical methods for PDEs, finite element methods, multigrid solvers, and high-performance computing. Current research integrates operator learning with multigrid PDE solvers, with broader interests in numerical analysis, scientific machine learning, and scalable computational methods.

Education

Ruhr University Bochum, Germany	Oct 2023–Present
M.Sc. Computational Engineering	Current GPA: 2.0
Hohai University, China	Sep 2012–Jul 2016
B.Eng. Engineering Mechanics	GPA: 82.03/100

Research Experience

Parallel Deep Operator Network as a Surrogate Component for Large-Scale Multigrid PDE Solvers

Supervisor: Prof. Dr. Martin Kronbichler

Jun.2026–Present

- Developed a surface finite element solver in `deal.II` for variable-coefficient Laplace–Beltrami problems on curved manifolds, including geometric multigrid, adaptive mesh refinement, and discretization-error analysis.
- Developed and evaluated DeepONet-based residual-to-correction operators as learned smoothers within multigrid, benchmarking them against classical Jacobi and Gauss–Seidel/SSOR smoothers using two-grid and V-cycle convergence.
- Analyzed the relationship between neural training objectives and numerical solver performance, including energy-error reduction, coarse-grid interaction, and generalization across mesh resolutions; investigated hybrid learned–classical smoothing strategies for improved robustness.

Project Experience

MPI-Based Molecular Dynamics Solver

Apr 2025–Jul 2025

- Developed a parallel Molecular Dynamics solver in C++ for large-scale particle systems governed by the Lennard–Jones potential, using the Velocity-Verlet method for numerical time integration.
- Implemented cell lists and Verlet neighbor lists for efficient evaluation of short-range interactions, reducing the computational complexity from $O(N^2)$ to approximately $O(N)$ under a fixed interaction cutoff.
- Parallelized the solver using MPI-based domain decomposition and analyzed strong and weak scaling, with emphasis on computational complexity, communication cost, and parallel efficiency.

Quantum–Classical Hybrid Computing Package

Dec 2025–Apr 2026

- Implemented HHL and Variational Quantum Linear Solver (VQLS) algorithms in C++ for small-scale linear systems $Ax = b$ using quantum simulation.
- Used the XACC framework to investigate intermediate representations (IR) for quantum–classical workflows, providing a basis for heterogeneous computation across classical and quantum computing backends.
- Investigated the distinction between classical linear-system solutions and their quantum-state representation, where the solution is encoded as the normalized state $|x\rangle = x/\|x\|$, highlighting limitations in recovering the full classical solution from quantum linear solvers.

Uncertainty Quantification with AI-Based Surrogate Models

Oct 2025–Jan 2026

- Constructed a neural-network surrogate in PyTorch to approximate the parameter-to-response map of a nonlinear finite element model with uncertain material parameters.
- Applied Monte Carlo sampling to the surrogate model for efficient uncertainty propagation, analyzing the dependence of structural stress responses on Young's modulus and yield stress.
- Evaluated the surrogate-based UQ framework through prediction accuracy and reliability quantities, including stress distributions and failure probabilities defined by critical response thresholds.

Nonlinear FEM Solver in Java

Aug 2025–Sep 2025

- Built a modular object-oriented FEM engine with Newton–Raphson and arc-length solvers for nonlinear systems, emphasizing algorithmic stability and memory-efficient sparse matrix operations.

Professional Experience

Xiamen Cardiovascular Hospital, Xiamen University

Jul 2017 – Sep 2023

Computational Engineer / Project Team Lead

Xiamen, China

- *Cardiovascular Devices*: Worked extensively with cardiovascular interventional devices, including coronary stents, angioplasty balloons, guidewires, and pacemakers, addressing practical challenges in device selection, specifications, clinical use, deployment constraints, and post-procedure risk management.
- *Computational Mechanics*: Applied Abaqus-based finite element simulations to investigate the deformation and mechanical response of medical devices, relating material properties, stress-strain behavior, geometry, and loading conditions to practical questions of structural performance and failure.
- *Clinical Risk Management*: Managed clinical risk for high-risk Class III implantable medical devices by investigating adverse events, analyzing clinical and device data, coordinating root-cause investigations with manufacturers, and preparing formal investigation reports; established standardized operating procedures (SOPs) for device traceability, risk management, and information handling.
- *Centralized Procurement*: Led hospital-level implementation of national centralized procurement programs for coronary stents (2019), angioplasty balloons (2021), and cardiac pacemakers (2022), translating policy changes into operational requirements across clinical demand, supplier coordination, reimbursement, and supply-chain management.
- *Supply-Chain Analytics*: Developed inventory monitoring and decision-support logic for specification-dependent medical-device demand, combining real-time stock levels with one-, three-, and six-month average consumption to forecast demand, guide procurement, and reduce the risks of shortages and excess inventory.
- *Decision Support & Budgeting*: Served as Secretary of the Medical Device Information Management Committee, analyzing monthly changes in device utilization, procurement, and healthcare policy; prepared annual procurement and budget plans and delivered data-driven reports to hospital leadership for operational and strategic decision-making.
- *Information Systems*: Led cross-functional information-system projects involving more than 40 medical-device suppliers and multiple clinical, technical, financial, and administrative departments; redesigned reimbursement data mappings and developed middleware APIs to integrate heterogeneous hospital systems, and contributed to peer-reviewed research on medical-device information management and system interoperability.

Xiamen International Bank, Corp. Ltd.

Jul 2016 – Jun 2017

Junior Banking Analyst

Xiamen, China

- *Financial & Risk Analysis*: Assessed corporate creditworthiness and repayment capacity through financial-statement analysis, cash-flow assessment, collateral valuation, and transaction data, supporting lending and risk-management decisions.
- *Interbank Market Operations*: Supported interbank lending and short-term funding transactions, analyzing liquidity requirements, counterparty conditions, and transaction terms to support funding and risk decisions.

Technical Skills

- *Programming & Numerical Computing*: C++, Python, Java, SQL, deal.II, Abaqus, NumPy, pandas.
- *Parallel & Scientific Computing*: MPI, OpenMP, SLURM, Linux, Docker, Git/GitHub, Jupyter.
- *Machine Learning & Heterogeneous Computing*: PyTorch, scikit-learn, XACC, Qiskit.

Publications

Fu, L., Yang, F., & Shi, Y. (2023). *Exploration and Practice of a UDI-Based Management Model*. China Medical Device Information.

Contribution: Led the conceptual design and system architecture of a middleware solution integrating six heterogeneous hospital information systems and developed custom APIs to support cross-departmental data interoperability and consistency.

Awards & Honors

Best Employee of the Year, Xiamen Cardiovascular Hospital, Xiamen University

Dec 2022

Star of Innovation, Xiamen Cardiovascular Hospital, Xiamen University

Dec 2020

Junior Banker Elite Program, Xiamen International Bank Co., Ltd.

Mar 2017

Academic Progress Award and Scholarship, Hohai University

Nov 2014